GARDE

Executive summary—GARDE, the primary care team, and primary care patients

- GARDE is a population-based tool that automates the provision of healthcare services, like USPSTF-recommended cancer genetic testing. This helps your team provide high quality care *without* adding to their workload.
- GARDE population health algorithms can screen patients' family histories that clinicians have already collected in the electronic health record (EHR) to identify patients eligible for genetic testing.
- GARDE includes a chatbot authoring platform (GARDE-Chat) for patient outreach, education and healthcare service facilitation.
- Recommended healthcare can be facilitated and centrally managed by genetic counselors or other clinicians.
- Identifying an individual with hereditary breast and ovarian cancer (HBOC) or Lynch syndrome can have a significant health impact because those individuals are at increased risk (up to 61%) to develop a wide spectrum of cancers. Cancer risks can be managed through enhanced cancer screening and/or preventive surgeries.

> "Using GARDE helped our health system offer evidence-based cancer genetic testing and expert interpretation to our patients without adding to the primary care workload."
>
> -- Medical Director for Quality, Division of Family and Community Medicine at an implementing site

What are some of the lifetime cancer risks associated with hereditary cancer syndromes?

Sex	**Cancer syndrome**	**Cancer site**	**Lifetime Risk**
Female	*BRCA1* or *BRCA2*	Breast	45 – 65%
Male	*BRCA1* or *BRCA2*	Prostate	7 – 61%
Both	*BRCA1* or *BRCA2*	Pancreas	5 – 10%
Both	Lynch syndrome	Colorectal	9 – 61%

What if a patient's family history in the EHR is incomplete?

GARDE does not require a complete pedigree. A large number of eligible patients can be identified using available family history in the EHR. At four institutions that implemented GARDE, 29,301 out of 530,308 patients screened (~5.5%) were identified as eligible for genetic testing.

How do patients pay for genetic testing?

For the vast majority of insured patients, the recommended genetic testing is a covered benefit and their out of pocket costs are typically <$100. Patients with limited resources can apply for support through laboratory patient assistance programs.

Can I trust GARDE?

GARDE was developed by an experienced biomedical informatics research team at University of Utah Health (UHealth) supported by the National Cancer Institute (NCI, U24CA204800). NCI supported a successful trial of GARDE at UHealth and New York University Langone Health (U01CA232826). With NCI support (U24CA274582), GARDE is being implemented at Medical University of South Carolina and Weill Cornell Medicine. Peer-reviewed publications about GARDE can be found here.

Can GARDE address other research questions or clinical needs?

GARDE can be used in both research and clinical settings. Additional use cases can be found here.

How many patients are likely to be tested through GARDE?

The below figure provides study outcomes of the BRIDGE trial at UofU Health and NYU Langone.

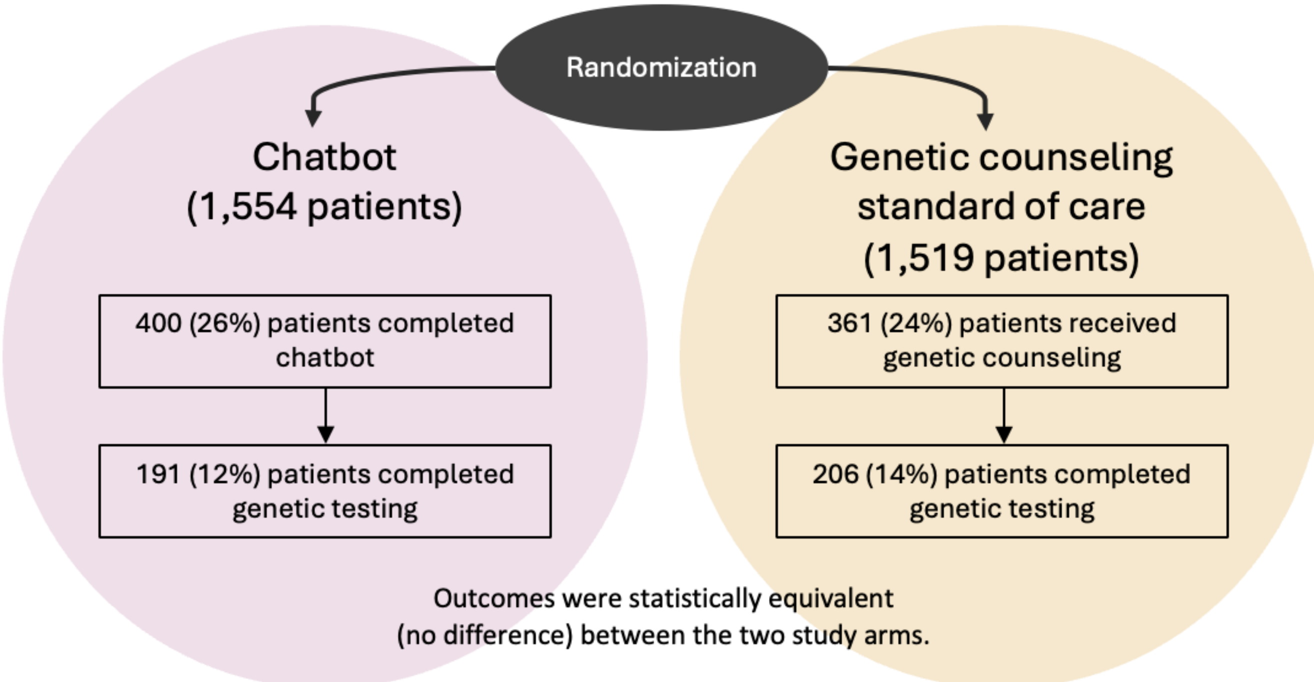

Figure 1 Uptake of Cancer Genetic Services BRIDGE study flow. Adapted from: Kaphingst KA, Kohlmann WK, Lorenz Chambers R, et al. Uptake of Cancer Genetic Services for Chatbot vs Standard-of-Care Delivery Models: The BRIDGE Randomized Clinical Trial. JAMA Netw Open*. 2024. PMID: 39250153; PMCID: PMC11385050.*

I have other questions about GARDE. Is there somebody I can speak with?

The GARDE study team can be reached at GARDE@hci.utah.edu and can provide you with references from implementing sites.